

Exam Date & Time: 15-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: ECE 2124- DIGITAL SYSTEM DESIGN

DIGITAL SYSTEM DESIGN [ECE 2124]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) Implement the logic function using an 8x1 MUX
 $F(A, B, C, D) = \sum m(1, 3, 4, 11, 12, 13, 14, 15)$ (4)
- 1B) Design a full subtractor using a Decoder and logic gates. Explain its working principle (3)
- 1C) Implement a two-input NAND gate with Transistor Transistor Logic (TTL) (3)
- 2A) Design a code converter to convert a 3 bit binary to gray code (4)
- 2B) Design a Mod- 10 Ripple up counter using T Flipflop (3)
- 2C) Derive the characteristic equation of SR flipflop. Convert SR flipflop to D flipflop (3)
- 3A) Design a sequence detector to detect 101 with overlapping using Melay model with positive edge triggered D flip flop (4)
- 3B) Design synchronous mod 8 up counter using T flip flop (3)
- 3C) Write dataflow Verilog code for the 2:4 decoder (3)
- 4A) Write the Structural Verilog code for 4 bit parallel adder (4)
- 4B) Write the Sequential verilog code for positive edge triggered JK Flipflop (3)
- 4C) Write the Verilog code for a 3 input XOR gate as a user defined primitive (UDP) (3)
- 5A) Implement 3-bit counter using ACT-3(S-module) FPGA. The logic expression of the counter are as follows (4)

$$D_1 = \overline{Q_1}, D_2 = Q_1 \oplus Q_2, D_3 = Q_1 Q_2 + Q_3(\overline{Q_1} + \overline{Q_2})$$

- 5B) Implement 2-bit magnitude comparator using Xilinx FPGA (3)
- 5C) Explain Y-Chart. (3)

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